ATM: Adaptive Arbitrary-Length Tokenization with Semantic Modeling for Time Series Forecasting

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Abstract

Recent advances in time series forecasting have achieved remarkable success, yet two critical challenges are not fully addressed: the limitations of fixed-length patching strategies and the lack of semantic-level modeling. Fixed-length patches struggle to capture heterogeneous temporal patterns and often truncate temporal patterns, while existing methods largely rely on data-driven statistical patterns without semantic guidance. We propose Adaptive Arbitrary-Length Tokenization with Semantic Modeling (ATM), a novel framework designed to address these issues. ATM introduces a Temporal Tokenization Module, which consists of two interrelated components: the Time Tokenizer, which adaptively partitions time series into arbitrary-length patches to preserve meaningful patterns (e.g., full cycles or peaks), and the Semantic Tokenization Regularization, designed to ensure semantically coherent temporal partitioning. In addition, ATM incorporates Semantic-Aware Modeling, where a Semantic Extractor enriches patches with latent semantic information and a Semantic Modeler captures hierarchical dependencies from local temporal patches to global sequence structures. Extensive experiments show that ATM consistently surpasses state-of-the-art methods in both long-term and short-term forecasting, demonstrating its effectiveness and strong generalization ability. Code is available at https://anonymous.4open.science/r/ATM.

CCS Concepts

• **Computing methodologies** → **Machine learning**.

Keywords

Time series forecasting; Adaptive tokenization; Semantic modeling

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1 Introduction

In recent years, with the rapid development of deep learning techniques, a wide range of time series forecasting models have been proposed [3, 6, 8, 16, 24, 26, 32]. Among them, patch-based methods [4, 13, 14, 33] and Transformer-based [19, 22, 25, 30] architectures have been widely applied. The former enhances the model's ability to capture local structural patterns in a high-dimensional representation space [2, 12]. By dividing time series into multiple patches, the model can extract richer temporal dependencies than those captured at individual time steps [20]. The latter [28], owing to its powerful semantic modeling capabilities, has achieved remarkable success in natural language processing [1] and computer vision [5], and has been extensively adapted for time series modeling tasks [31].

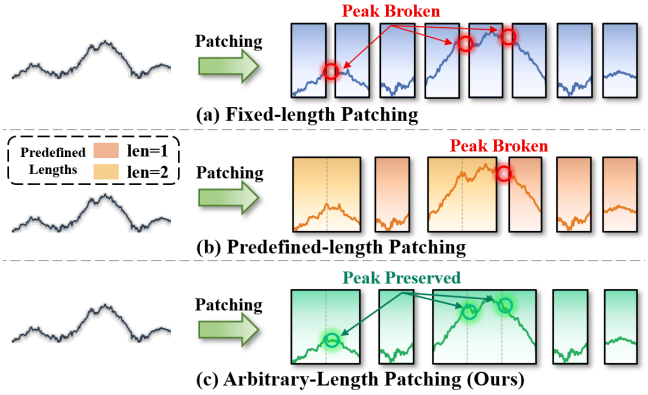


Figure 1: Comparison of existing patching methods, highlighting the ability of our method to preserve temporal patterns.

Table 1: Efficiency comparison of different models

Model	Parameters (M)	Training Step Time (ms)	Inference Step Time (ms)
TimesNet	1.71	132.40	30.34
HDMixer	1.05	80.86	24.79
DualSG	2.14	136.96	32.09
ATM	1.58	124.32	26.91

Despite the outstanding performance of existing methods [10, 17], two critical issues are not fully addressed: (1) the limitations of fixed-length patching strategies and (2) the lack of semantic modeling for time series data.

1.0.1 Implementation Details. Our model is optimized with Adam optimizer [9]. To ensure fair comparison, following the settings in TFB [23] and FoundTS [11], we do not apply the "Drop Last" trick. For long-term forecasting tasks, the L1 loss is employed across all datasets. For short-term forecasting, the model is refined using the SMAPE loss. All our training processes are conducted using PyTorch [21] in Python 3.11 and executed on a single RTX 4090 GPU.

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Table 2: Average performance comparison between ATM and HDMixer. The best performance is highlighted in red, and the second-best is underlined. Statistical significance is evaluated by paired t-test.

Models		ATM		HDMixer		Significance
Metric		MSE	MAE	MSE	MAE	p-value
ETTh1	Avg	0.426 ± 0.003	**0.422 ± 0.002**	0.435 ± 0.004	0.430 ± 0.003	0.021
ETTh2	Avg	0.358 ± 0.002	**0.384 ± 0.003**	0.351 ± 0.002	0.384 ± 0.002	0.023
ETTm1	Avg	**0.379 ± 0.003**	**0.382 ± 0.003**	0.389 ± 0.003	0.389 ± 0.004	0.019
ETTm2	Avg	**0.277 ± 0.002**	**0.318 ± 0.004**	0.282 ± 0.003	0.325 ± 0.003	0.024
Traffic	Avg	0.442 ± 0.004	**0.275 ± 0.002**	0.424 ± 0.003	0.280 ± 0.002	0.045
Weather	Avg	**0.249 ± 0.003**	**0.266 ± 0.003**	0.254 ± 0.003	0.270 ± 0.004	0.030
Electricity	Avg	0.181 ± 0.002	**0.265 ± 0.003**	0.177 ± 0.002	0.272 ± 0.003	0.041

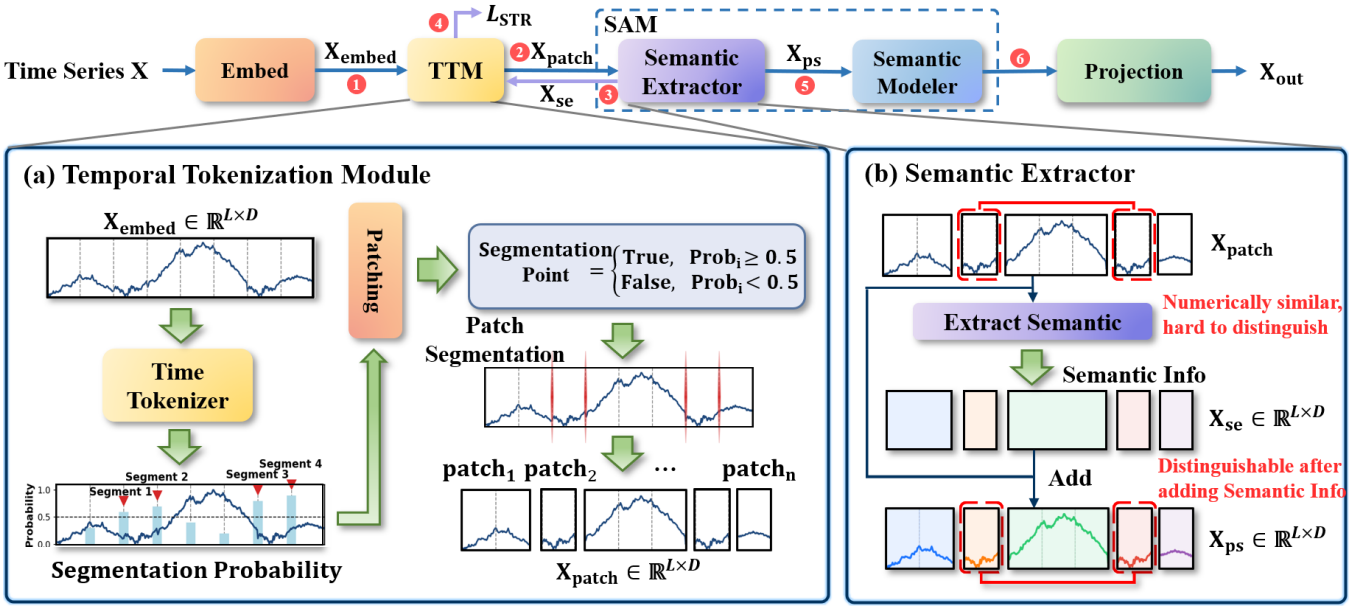


Figure 2: Overall structure and data flow of the proposed ATM. Temporal Tokenization Module (TTM) adaptively partitions time series into arbitrary-length. Semantic Extractor enriches patches with semantic information for disambiguating superficially similar patterns.

Table 3: Comparison of existing patch-based methods.

Model	Adaptive Number of Patches	Arbitrary Patch Lengths	No Predefined Patch Length Candidates	Preserve Complete Temporal Patterns
PatchTST [20]	✗	✗	✗	✗
PatchMLP [27]	✓	✗	✗	Partial ✓
DualSG [4]	✓	✗	✗	Partial ✓
APN [15]	✗	✓	✓	Partial ✓
HDMixer [7]	✗	✗	✓	Partial ✓
ATM	✓	✓	✓	✓

1.1 Long-term Forecasting

1.1.1 Setups. We conduct experiments on seven widely-used real-world datasets, including four ETT datasets (ETTh1, ETTh2, ETTm1, ETTm2), Traffic, Weather, and Electricity. The input time series length L is fixed as 96 for a fair comparison, and we adopt four distinct prediction horizons $H \in \{96, 192, 336, 720\}$. Consistent with prior works [29], the mean squared error (MSE) and mean absolute error (MAE) are employed as evaluation metrics.

1.1.2 Results. The average results of long-term forecasting are presented in Table 4. ATM demonstrates state-of-the-art performance, achieving the best results in 51 evaluation tasks, whereas the closest competing baseline achieves the best results in only 6 tasks. Notably, ATM achieves an average reduction of 4.66% in MSE and 4.08% in MAE compared to the state-of-the-art patch-based model PatchTST on the ETTh1 dataset, which demonstrates that adaptive patching is able to preserve temporal pattern information more effectively than fixed-length patching. Compared with Transformer-based baselines, ATM achieves a 3.99% reduction in MSE on the ETTh2 dataset relative to TQNet, and yields reductions

Table 4: The full results of long-term multivariate forecasting. The best performance is highlighted in red, and the second-best is underlined.

Models		**ATM**		Times2D		TQNet		iTransformer		PatchTST		CALF		DualSG		APN		AMD		HDMixer		TimesNet		MICN	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	**0.369**	**0.384**	0.371	0.409	0.372	0.391	0.386	0.404	0.394	0.407	0.370	0.388	0.370	0.387	0.379	0.390	0.370	0.397	0.372	0.398	0.383	0.403	0.418	0.429
	192	0.423	**0.415**	0.430	0.427	0.430	0.421	0.441	0.436	0.446	0.434	0.427	0.421	**0.420**	0.418	0.432	0.425	0.431	0.418	0.433	0.422	0.438	0.430	0.470	0.485
	336	0.456	**0.436**	0.464	0.439	0.475	0.438	0.486	0.457	0.485	0.451	0.455	0.436	**0.440**	0.436	0.465	0.441	0.462	0.438	0.465	0.439	0.491	0.467	0.570	0.551
	720	**0.457**	**0.454**	0.469	0.460	0.481	0.469	0.508	0.493	0.479	0.471	0.475	0.467	0.470	0.459	0.474	0.470	0.464	0.463	0.468	0.460	0.520	0.498	0.769	0.671
	Avg	0.426	**0.422**	0.434	0.434	0.440	0.430	0.455	0.448	0.451	0.441	0.432	0.428	**0.425**	0.425	0.438	0.432	0.432	0.429	0.435	0.430	0.458	0.450	0.557	0.534
ETTh2	96	**0.275**	**0.323**	0.284	0.334	0.294	0.341	0.300	0.347	0.294	0.342	0.279	0.330	0.283	0.332	0.290	0.335	0.279	0.331	0.280	0.331	0.341	0.374	0.298	0.364
	192	**0.355**	**0.376**	0.360	0.379	0.365	0.390	0.379	0.398	0.378	0.395	0.357	0.381	0.358	0.379	0.371	0.387	0.358	0.383	0.360	0.381	0.400	0.411	0.440	0.454
	336	0.402	0.412	0.387	0.419	0.415	0.426	0.418	0.429	0.381	0.409	0.362	0.399	0.383	0.405	0.408	0.416	0.407	0.411	**0.360**	**0.395**	0.453	0.450	0.655	0.565
	720	**0.400**	**0.426**	0.410	0.428	0.431	0.442	0.428	0.443	0.412	0.433	0.404	0.427	0.411	0.435	0.411	0.432	0.404	0.431	0.404	0.429	0.460	0.467	0.949	0.713
	Avg	0.358	**0.384**	0.360	0.390	0.376	0.400	0.381	0.404	0.366	0.395	**0.351**	0.384	0.359	0.388	0.370	0.393	0.362	0.389	0.351	0.384	0.414	0.426	0.586	0.524
ETTm1	96	**0.311**	**0.341**	0.325	0.350	0.323	0.349	0.339	0.370	0.321	0.360	0.321	0.349	0.320	0.349	0.320	0.356	0.324	0.352	0.322	0.351	0.335	0.374	0.319	0.363
	192	**0.362**	**0.370**	0.371	0.376	0.371	0.374	0.382	0.395	0.368	0.384	0.373	0.375	0.367	0.374	0.368	0.385	0.371	0.376	0.368	0.379	0.374	0.387	0.370	0.390
	336	**0.391**	**0.392**	0.408	0.400	0.405	0.399	0.418	0.419	0.399	0.402	0.407	0.398	0.397	0.394	0.398	0.401	0.406	0.398	0.403	0.395	0.411	0.412	0.406	0.425
	720	**0.451**	**0.427**	0.470	0.432	0.460	0.432	0.487	0.456	0.453	0.435	0.476	0.437	0.452	0.428	0.453	0.434	0.462	0.431	0.461	0.430	0.478	0.448	0.480	0.477
	Avg	**0.379**	**0.382**	0.394	0.390	0.390	0.389	0.407	0.410	0.385	0.395	0.394	0.390	0.384	0.387	0.385	0.394	0.391	0.389	0.389	0.389	0.400	0.405	0.394	0.414
ETTm2	96	**0.171**	**0.249**	0.181	0.257	0.179	0.255	0.185	0.272	0.179	0.260	0.178	0.255	0.175	0.252	0.179	0.258	0.178	0.256	0.177	0.255	0.186	0.265	0.180	0.275
	192	**0.237**	**0.295**	0.242	0.300	0.245	0.298	0.253	0.313	0.249	0.306	0.241	0.297	0.240	0.296	0.247	0.302	0.242	0.297	0.243	0.301	0.258	0.310	0.306	0.374
	336	**0.299**	**0.334**	0.308	0.341	0.306	0.339	0.315	0.350	0.312	0.346	0.305	0.339	0.304	0.339	0.310	0.343	0.307	0.338	0.303	0.340	0.318	0.351	0.324	0.388
	720	**0.400**	**0.393**	0.403	0.395	0.404	0.395	0.413	0.406	0.405	0.396	0.403	0.396	0.403	0.395	0.406	0.398	0.404	0.397	0.404	0.402	0.408	0.403	0.500	0.485
	Avg	**0.277**	**0.318**	0.284	0.323	0.284	0.322	0.292	0.335	0.286	0.327	0.282	0.322	0.281	0.321	0.286	0.325	0.283	0.322	0.282	0.325	0.293	0.332	0.327	0.381
Traffic	96	0.410	0.261	0.407	0.260	0.410	0.261	0.398	0.269	0.520	0.315	0.406	0.268	0.403	0.265	0.422	0.284	**0.396**	0.265	0.399	**0.259**	0.582	0.318	0.518	0.309
	192	0.429	**0.272**	0.428	0.279	0.429	0.277	0.421	0.277	0.492	0.309	0.430	0.278	0.397	0.277	0.452	0.299	0.398	0.278	**0.396**	0.277	0.616	0.332	0.537	0.315
	336	0.446	**0.274**	0.445	0.284	0.451	0.282	0.439	0.282	0.504	0.311	0.445	0.281	0.442	0.281	0.465	0.289	0.441	0.280	**0.438**	0.283	0.625	0.334	0.533	0.312
	720	0.481	**0.291**	0.481	0.301	0.484	0.298	0.469	0.301	0.518	0.315	0.477	0.300	0.439	0.300	0.501	0.310	**0.438**	0.298	0.461	0.300	0.639	0.350	0.577	0.324
	Avg	0.442	**0.275**	0.440	0.281	0.444	0.280	0.432	0.282	0.509	0.313	0.440	0.282	0.420	0.281	0.460	0.296	**0.418**	0.280	0.424	0.280	0.616	0.334	0.541	0.315
Weather	96	**0.164**	**0.200**	0.168	0.205	0.165	0.204	0.173	0.214	0.176	0.218	0.165	0.204	0.166	0.205	0.174	0.212	0.167	0.203	0.167	0.203	0.171	0.220	0.166	0.229
	192	**0.213**	**0.243**	0.216	0.255	0.216	0.247	0.221	0.254	0.223	0.257	0.215	0.250	0.217	0.246	0.221	0.253	0.217	0.248	0.219	0.247	0.216	0.261	0.220	0.278
	336	0.269	**0.286**	0.271	0.292	0.272	0.289	0.278	0.296	0.278	0.297	0.270	0.290	0.268	0.288	0.279	0.294	**0.265**	0.293	0.273	0.289	0.281	0.305	0.275	0.331
	720	0.349	**0.336**	0.356	0.355	0.355	0.342	0.358	0.346	0.354	0.347	0.355	0.350	0.353	0.341	0.358	0.349	0.355	0.350	0.355	0.341	0.364	0.360	**0.344**	0.355
	Avg	**0.249**	**0.266**	0.253	0.277	0.252	0.271	0.258	0.278	0.258	0.280	0.252	0.274	0.251	0.270	0.258	0.277	0.251	0.274	0.254	0.270	0.258	0.287	0.251	0.298
Electricity	96	0.160	0.250	0.164	0.245	**0.134**	0.237	0.148	0.240	0.194	0.286	0.145	0.238	0.135	0.236	0.181	0.280	0.139	0.236	0.146	**0.235**	0.168	0.270	0.166	0.268
	192	0.169	**0.252**	0.171	0.267	**0.154**	0.261	0.162	0.267	0.199	0.291	0.161	0.261	0.155	0.260	0.185	0.288	0.158	0.270	0.164	0.273	0.184	0.290	0.174	0.287
	336	0.180	**0.265**	0.179	0.273	0.171	0.273	0.178	0.275	0.215	0.308	0.175	0.272	0.172	0.273	0.201	0.296	**0.167**	0.273	0.174	0.274	0.198	0.298	0.190	0.306
	720	**0.213**	**0.294**	0.227	0.315	0.223	0.305	0.225	0.317	0.256	0.332	0.222	0.304	0.223	0.306	0.241	0.320	0.224	0.305	0.223	0.305	0.221	0.319	0.223	0.321
	Avg	0.181	**0.265**	0.185	0.275	**0.171**	0.269	0.178	0.275	0.216	0.304	0.176	0.269	0.171	0.269	0.202	0.296	0.172	0.271	0.177	0.272	0.193	0.294	0.188	0.296
1st Count		**51**		0		3		0		0		1		3		0		5		6		0		1	

of 5.51% in MSE and 5.58% in MAE on the ETTh1 dataset relative to iTransformer.

1.2 Short-term Forecasting

1.2.1 Setups. We employ the M4 datasets [18], comprising 100, 000 univariate marketing data with six frequencies from hourly to yearly. In this setting, the prediction horizons are relatively short, ranging in [6, 48]. Accordingly, the input lengths are configured to be twice the size of prediction horizons. The evaluation metrics are symmetric mean absolute percentage error (SMAPE), mean absolute scaled error (MASE), and overall weighted average (OWA).

1.2.2 Results. As shown in Table 5, ATM demonstrates superior performance in short-term forecasting across multiple evaluation metrics. Notably, it achieves the best results in 12 out of 15 categories, substantially outperforming all baselines. In comparison with PatchTST, the state-of-the-art patch-based method, ATM achieves

Table 5: The full results of short-term forecasting on M4 dataset. The input length and prediction length are set to [12, 96] and [6, 48], respectively.

Models		**ATM**	Times2D	PatchTST	ETSformer	FEDformer	GPT4TS	TimeLLM	DLinear	TiDE	TCN	MICN
Yearly	SMAPE	**13.331**	13.416	13.476	18.007	13.727	13.530	13.418	16.966	15.318	14.918	25.020
	MASE	**2.991**	2.992	3.019	4.487	3.046	3.010	3.005	4.285	3.539	3.364	7.162
	OWA	**0.784**	0.785	0.791	1.115	0.803	0.790	0.788	1.058	0.910	0.879	1.668
Quarterly	SMAPE	10.267	10.364	10.382	13.376	10.793	10.178	**10.118**	12.144	11.831	11.120	15.217
	MASE	1.205	1.222	1.234	1.908	1.284	1.193	**1.188**	1.521	1.410	1.361	1.963
	OWA	**0.904**	0.915	0.923	1.305	0.958	0.906	0.905	1.107	1.052	1.001	1.407
Monthly	SMAPE	**12.885**	12.920	12.961	14.561	14.262	12.894	12.982	13.515	15.183	15.625	16.945
	MASE	**0.975**	0.979	0.975	1.370	1.105	0.976	0.978	1.037	1.191	1.274	1.443
	OWA	**0.905**	0.906	0.906	1.152	1.014	0.906	0.905	0.956	1.090	1.141	1.265
Others	SMAPE	**4.938**	4.950	4.955	7.269	4.957	4.942	4.939	6.713	6.120	7.181	41.985
	MASE	**3.320**	3.342	3.349	5.244	3.367	3.322	3.321	4.955	4.330	4.679	62.734
	OWA	1.040	1.047	1.048	1.593	1.039	1.025	**1.005**	1.488	1.330	1.494	14.313
Average	SMAPE	**11.962**	12.022	12.060	14.705	12.841	11.993	11.993	13.640	13.956	13.959	19.640
	MASE	**1.611**	1.618	1.626	2.410	1.707	1.613	1.612	2.096	1.941	1.946	5.948
	OWA	**0.860**	0.865	0.870	1.172	0.919	0.863	0.860	1.052	1.021	1.023	2.279
1st Count		**12**	0	0	0	0	0	3	0	0	0	0

an overall improvement of 1% in SMAPE. Compared with Transformer-based baselines, ATM achieves a 0.9% reduction in SMAPE relative to Times2D, and yields a reduction of 18.68% in SMAPE relative to ETSformer.

1.3 Semantic Clustering Visualization

To validate the ATM's capability to distinguish time-series patterns that exhibit numerical resemblance but semantic divergence, we visualize the patch-level semantic clustering results of STR.

Figure 3 illustrates how our method captures and differentiates patch-level semantic patterns despite numerical resemblance. In Figure 3 (a), patches are clustered according to their learned semantic representations, showing that semantically similar patterns are grouped together. In Figure 3 (b), we highlight a pair of sequences (circled in red in (a)) that are numerically similar but assigned to different clusters, indicating that ATM captures semantic differences. In Figure 3 (c), a pair of similar sequences (circled in green in (a)) is assigned to the same cluster, demonstrating that the model successfully identifies shared semantic features. Overall, these visualizations confirm that our method captures semantic distinctions beyond numerical similarity at the patch level.

1.4 Ablation Study

We conduct ablation studies on three datasets by removing (w/o) specific modules from ATM.

(1) w/o-Time Tokenizer: Time Tokenizer removed.

(2) w/o-STR: Semantic Tokenization Regularization removed.

(3) w/o- Semantic Extractor: Semantic Extractor removed.

The results reported in Table 6 demonstrate that removing the Time Tokenizer leads to a substantial performance reduction, with

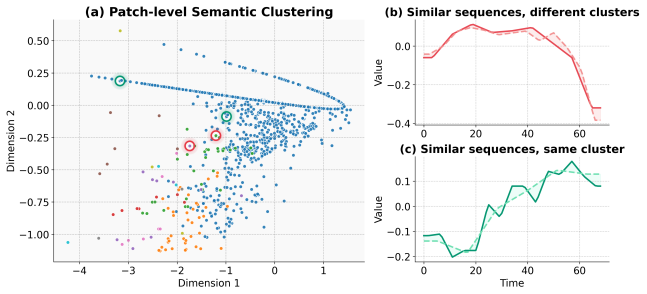


Figure 3: Visualization of patch-level semantic clustering results of STR.

average increases of 2.32%/2.15% in MSE/MAE on ETTh1. Incorporating STR yields a 3.34%/3.64% improvement in MSE/MAE on ETTm2. Similarly, the addition of Semantic Extractor yields a 4.07%/4.76% improvement in MSE/MAE on Weather. The best overall performance is achieved when all three modules are jointly employed, achieving the lowest MSE and MAE across all datasets.

1.5 Hyperparameter Sensitivity

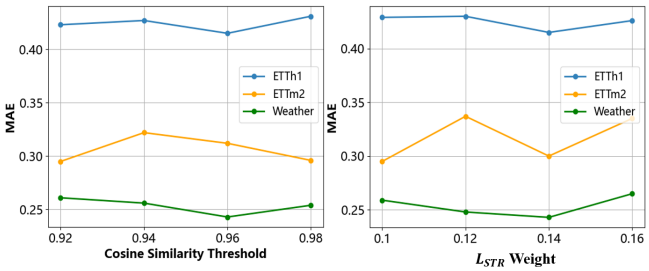
In this section, we investigate the key hyperparameters of ATM, including the number of convolutional layers H in the Time Tokenizer, the cosine similarity threshold r and the L_{STR} weight λ. The analysis is conducted on ETTh1, ETTm2, Weather datasets with a fixed forecasting horizon of 192, where only one hyperparameter is varied while the others remain fixed.

As shown in Figure 4, ATM achieves the best performance when both the cosine similarity threshold and the STR weight are set to intermediate values. This observation indicates that, for both

Table 6: Ablation on different modules on ETTh1, ETTm2 and Weather datasets.

Design	Time Tokenizer	STR	Semantic Extractor	ETTh1		ETTm2		Weather	
				MSE	MAE	MSE	MAE	MSE	MAE
ATM	✓	✓	✓	**0.426**	**0.422**	**0.277**	**0.318**	**0.249**	**0.266**
w/o	w/o	✓	✓	0.435	0.436	0.289	0.328	0.258	0.277
w/o	✓	w/o	✓	0.434	0.436	0.290	0.329	0.260	0.281
w/o	✓	✓	w/o	0.438	0.439	0.294	0.332	0.263	0.285

hyperparameters, moderate settings are more effective in capturing meaningful time-series representations. When the cosine similarity threshold is set too small or too large, the model becomes less effective at distinguishing patch-level semantic patterns, leading to degraded performance. Similarly, inappropriate values of the STR weight—either overly weak or overly strong—impair the learning of semantic representations and result in a gradual increase in the prediction loss.



(b) Cosine Similarity Threshold (c) L_{STR} Weight

Figure 4: Part of sensitivity analysis of key hyperparameters for ATM with a prediction horizon of 192. The full results are in Appendix B.

2 Conclusion

In this work, we address two key limitations in time series forecasting—the unreasonable fixed-length patching and the lack of semantic guidance—by proposing ATM. Specifically, ATM introduces a Temporal Tokenization Module (TTM), which leverages a Time Tokenizer to adaptively partition time series into arbitrary-length patches and a Semantic Tokenization Regularization to ensure semantically coherent patching. Furthermore, ATM incorporates the Semantic-Aware Modeling (SAM) to extract semantic information for disambiguating superficially similar patterns, along with hierarchical dependency learning that ranges from patch-level dependencies to sequence-level interactions. Extensive experiments show that ATM achieves state-of-the-art performance across a wide range of time series forecasting benchmarks, highlighting its effectiveness and strong generalization. In future work, we plan to extend ATM to a broader spectrum of tasks and real-world scenarios beyond time series forecasting. Moreover, we will further investigate its robustness and scalability in more complex and practical settings, including scenarios with higher dimensionality, longer sequences, and more challenging real-world constraints.

Ethical Considerations

Our work exclusively uses publicly available benchmark datasets that contain no personally identifiable information. No human subjects are involved in this research.

Reproducibility statement

The promise that all experimental results can be reproduced. We have released our model code in an anonymous repository: https://anonymous.4open.science/r/ATM.

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A Dataset Details

A.1 Long-term Forecasting

We evaluate our approach on seven commonly used time series datasets for long-term forecasting. Following the methodology outlined in [29], each dataset is divided chronologically into training, validation, and test sets. For the ETT datasets, we use a 6:2:2 split ratio, while a 7:1:2 ratio is applied to the remaining datasets. A comprehensive overview of these datasets is presented below:

(1) **ETT** (Electricity Transformer Temperature) dataset comprises measurements of transformer temperatures and power loads from two regions in China during 2016–2018. It is available at two temporal resolutions: **ETTh** (hourly) and **ETTm** (15-minute intervals).
(2) **Traffic** dataset documents hourly road occupancy rates from 862 sensors installed on freeways in the San Francisco Bay Area. Spanning 2015–2016, it offers comprehensive insights into traffic conditions and congestion trends.
(3) **Weather** dataset captures 21 meteorological variables across Germany at 10-minute intervals throughout the year 2020. Key measurements include air temperature, visibility, among others, providing a detailed representation of weather variations.
(4) **Electricity** dataset records hourly electricity consumption (in kWh) for 321 individual clients. Collected from the UCI Machine Learning Repository between 2012 and 2014, it reflects the consumption patterns of households and businesses.

Detailed statistics for these datasets are shown in Table 7.

Table 7: The statistics of long-term forecasting datasets.

Datasets	Time steps	Channels	Frequency
ETTh1	17420	7	1 hour
ETTh2	17420	7	1 hour
ETTm1	69680	7	15 min
ETTm2	69680	7	15 min
Traffic	17544	862	1 hour
Weather	52696	21	10 min
Electricity	26304	321	1 hour

A.2 Short-term Forecasting

The M4 benchmark comprises a large collection of 100,000 time series drawn from diverse domains related to business, finance, and economic forecasting. The dataset is divided into six subsets, each corresponding to a different sampling frequency ranging from yearly to hourly, thereby covering a wide spectrum of temporal granularities encountered in real-world forecasting tasks. These subsets include series with varying lengths, patterns, and levels of complexity, posing diverse challenges for forecasting models. The detailed characteristics of the M4 datasets, including the number of time steps and domain categories, are summarized in Table 8.

Table 8: The statistics of short-term forecasting datasets.

Datasets	Time steps	Frequency	Domains
M4-Yearly	23000	Yearly	Demographic
M4-Quarterly	24000	Quarterly	Finance
M4-Monthly	48000	Monthly	Industry
M4-Weekly	359	Weekly	Macro
M4-Daily	4227	Daily	Micro
M4-Hourly	414	Hourly	Other

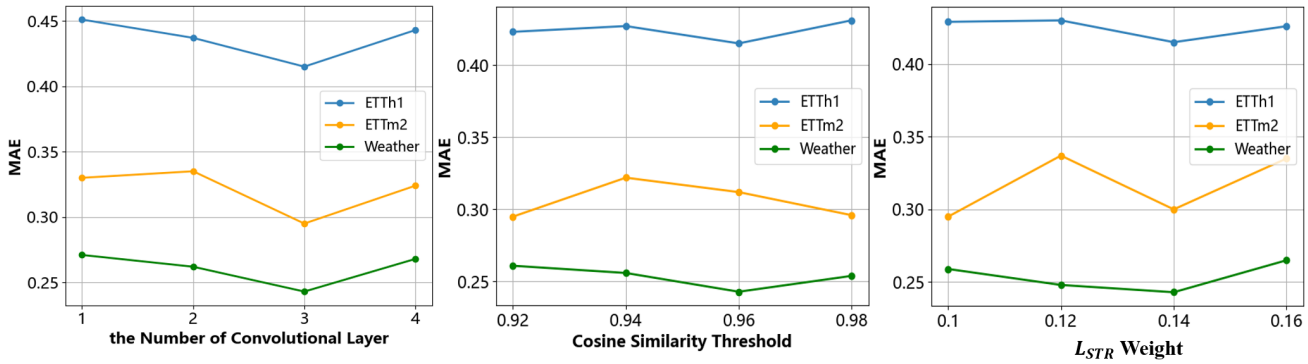
B Hyperparameter Sensitivity

The full hyperparameter sensitivity results are presented in this section. We investigate several key hyperparameters of ATM, including the number of convolutional layers H in the Time Tokenizer, the cosine similarity threshold r, and the weight λ of the L_{STR} regularization term. All experiments are conducted on the ETTh1, ETTm2, Weather datasets with a fixed forecasting horizon of 192. During the sensitivity analysis, only one hyperparameter is varied at a time, while the remaining hyperparameters are kept unchanged to ensure a fair comparison.

As illustrated in Figure 5, ATM achieves optimal performance when the number of convolutional layers is set to 3. When fewer convolutional layers are used, the MAE increases noticeably, suggesting that an insufficient model depth limits the ability of the Time Tokenizer to capture informative temporal patterns. Conversely, further increasing the number of layers does not lead to additional performance gains, indicating diminishing returns and potential over-parameterization.

In addition, the results show that intermediate values of both the cosine similarity threshold r and the STR loss weight λ yield better forecasting accuracy. Extremely small values weaken the effect of structural regularization, while overly large values impose excessive constraints on the model, both of which result in a gradual degradation in performance. These observations demonstrate that a balanced configuration of hyperparameters is crucial for achieving stable and robust performance with ATM.

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(a) the Number of Convolutional Layer (b) Cosine Similarity Threshold (c) L_{STR} Weight

Figure 5: Sensitivity analysis of key hyperparameters for ATM with a prediction horizon of 192.